

Request for Information (RFI)
Advancement of “Science as a Service” for NASA and Commercial Partners

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Short URL: <https://go.nasa.gov/saasrfi>

This Science Mission Directorate (SMD) Request for Information (RFI) does not constitute a commitment, implied or otherwise, that the National Aeronautics and Space Administration (NASA) will take action in this matter.

This RFI seeks responses from U.S. industry, universities, non-profit organizations, individuals, NASA centers, Federal Funded Research and Development Centers (FFRDCs) such as the Jet Propulsion Laboratory, and other U.S. federal, state, local, tribal government agencies as well as the international science community. Responses are subject to the restrictions placed on NASA by SEC. 526 of the “Consolidated Appropriations Act, 2024” that prohibit bilateral relationships with China. Details of that restriction are at:

<https://www.congress.gov/118/plaws/publ42/PLAW-118publ42.pdf> and at:

<https://science.nasa.gov/researchers/sara/faqs/prc-faq-roses/>

1. Summary

The Earth Science Technology Office (ESTO) in the Earth Science Division, the Heliophysics Space Weather Program in the Heliophysics Division (HPD), and NASA’s Astrophysics Division (APD) request industry input to inform the development of the Agency’s “Science as a Service (SaaS)” initiative. SaaS is designed to leverage existing capabilities of industry and identify where new technology investments by the Government will accelerate technology development. It will also enable the testing, validation, and rapid transition of next generation NASA technologies for operational use.

For ESTO, this RFI will inform the new Venture Technology Accelerator (VTA) program, which has been recently launched and builds upon the success of its legacy In-Space Validation of Earth Science Technologies (InVEST) program. While InVEST validated new technologies in space, VTA is designed to speed their transition to useful science and commercialization. This RFI is being used to help inform ESTO as to where new investment is needed, and to identify better processes for working with industry to achieve successful technologies in the commercial sector. It also will help to refine the concepts and priorities of the VTA program

(<https://esto.nasa.gov/vta>).

For the Heliophysics Space Weather Program, this RFI will inform strategies for the continued development of its Space Weather Observations Architecture. The goal is to identify pathways that encourage public-private partnerships, enabling collaboration between instrument developers and satellite constellation operators to proliferate Earth orbit with distributed monitoring stations.

For NASA's Astrophysics Division, this RFI seeks to identify emerging commercial capabilities for spacecraft buses, telescopes, and other observatory hardware, including up to entire observatories, to allow expansion of VTA-like approaches to the astrophysics domain. This includes information on repurposing, improving, or capabilities to meet the needs of space-based astrophysics observatories.

This RFI also seeks information on potential public-private astrophysics partnerships that would advance space-based astrophysics through collaboration or coordination with foundations and/or other private entities. Those partnerships could include co-investment in future observatories, or in expanding/extending the capabilities of current observatories. These partnerships should leverage the best capabilities and approaches from all partnering institutions. For NASA interests in future observatories, please see the co-released Community Announcement entitled "Mission Concepts for the Astrophysics Strategic Technology & Research Accelerator (ASTRA) Initiative."

This RFI specifically seeks commercial perspectives regarding:

- Technologies required to expand commercial business cases where NASA investments (validation, flight access, integration support) could reduce technical, schedule, or financial risk.
- Which NASA Earth Science, Space Weather, and astrophysics mission concepts or technology demonstrations could be attractive for commercial hosting, adoption, or downstream commercialization (e.g., instruments as product lines, data-as-a-service, analytics).
- Operational and integration needs to better align VTA and other technology development activities with commercial development cycles, platform interfaces, and investment strategies.

2. Background

In the Earth Science Division, VTA is an emerging ESTO initiative to mature high potential technologies—advanced sensors, instruments, subsystems, components, and -data processing systems—through -in-space demonstrations and -on-orbit- validation. For the Heliophysics Division, in alignment with the Space Weather Program's observation goals, this effort seeks to shift from traditional monolithic missions toward a resilient architecture of proliferated, hosted sensors. This effort also seeks to identify emerging capabilities or entities that would open up similar opportunities in Astrophysics. To better serve the growing commercial Earth -observation, space weather monitoring and space technology- market, SaaS aims to:

- Lower technical, financial, and schedule risks associated with commercializing and deploying new Earth-observing, space weather monitoring, and astrophysics technologies.
- Provide streamlined and predictable access to spaceflight launch opportunities aligned with commercial timelines.
- Support dual use development pathways beneficial to NASA science and viable- commercial markets.

- Facilitate rapid adoption by commercial, operational, and federal partners and identify VTA and other Agency demonstrations with strong commercialization potential (e.g., hosted payloads, instrument lines, data services).

This effort aligns with ongoing SMD goals to advance Earth system science, space weather resiliency, and innovative astrophysics approaches; foster partnerships; and deliver innovations that improve resource management, disaster response, climate resilience, protection of infrastructure, and national security.

3. Requested Response Topics

NASA seeks industry input addressing one or more of the following topics. Respondents may select a primary topic and up to three secondary topics (see Section 5).

Topic A: Commercial Business Case Technology Needs

Identify sensors, instruments, components, subsystems, or data processing innovations that commercial providers need to enable current or planned business cases within the next 3–5 years. Please specify:

- Target performance thresholds and validation metrics needed to unlock market entry, investor confidence, or customer acquisition.
- Gaps not served by the commercial supply chain (e.g., calibration standards, thermal-vac compatibility, rad-hard components, on-board processing).
- How VTA, the Space Weather Program, and the Astrophysics Division could reduce the development risk or reduce time-to-market (e.g., environmental testing, hosted on-orbit validation, reference datasets).

Topic B: Demonstration Pathways and On-Orbit Validation Needs for Commercial Use

Describe flight needs, testbed concepts, calibration/validation requirements, and operational constraints relevant to demonstrating commercial capabilities, including:

- Preferred platforms (rideshare, hosted payload, smallsat buses, International Space Station, commercial stations) and necessary interfaces.
- Validation data, documentation, and certification artifacts that would materially support business development and regulatory engagement.
- Opportunities for joint campaigns or co-investment- in demonstrations aligned with commercial schedules

Topic C: Streamlined Access to Space and Integration Support (Commercial Interfaces)

Recommend mechanisms, processes, or infrastructure needed to simplify commercial integration, including:

- Standardized payload interfaces, electrical/mechanical envelopes, and data links common in commercial fleets.
- Pre-negotiated rideshare/hosting options, rapid contracting pathways, and shared environmental test facilities.
- Ground segment and data operations support (e.g., cloud pipelines, metadata standards, application-programmer interface access) that enable commercial service delivery.

Topic D: Transition from Demonstration to Commercial Operations

Identify barriers and opportunities for transitioning NASA-validated technologies into commercial offerings, such as but not limited to:

- Productization (instrument lines, modules), manufacturability, supply chain, and quality assurance needs.
- Business models (data as-a-service, analytics, licensing) and the role of NASA validation in customer acquisition.
- Long-term lifecycle support needs (maintenance, calibration updates, re-flights, software upgrades).

Topic E: Partnerships and Community Ecosystem

Recommend approaches to strengthen NASA–commercial partnerships:

- Cofunded pilots, -milestone-based co--investment, or shared- risk- flight demonstrations.
- Collaboration frameworks with federal users (e.g., USDA, FEMA, NOAA, DOI) that also support commercial service uptake.

Topic F: Program Design, Funding Models, and Milestone Structures

Comment on programmatic elements such as:

- Pay for- -performance/milestone incentives and rapid decision cycles.
- Agile management approaches, phased demonstration gates, and prototyping pathways tailored to commercial cadence.
- Data sharing- frameworks, data rights, and licensing options that respect commercial IP while enabling scientific utility.

Topic G: Market-Driven- Application Needs

Describe commercial needs for actionable data/products in domains such as:

- Climate analytics, insurance/finance risk products
 - Agriculture, carbon markets, supply chain- monitoring
 - Water and resource management, infrastructure resilience
 - Maritime/aviation situational awareness, environmental compliance
 - Space weather resilience, including satellite drag modeling, radiation hazard assessment for orbital assets, and GNSS/PNT (Global Navigation Satellite Systems / Positioning, Navigation, and Timing) accuracy for critical infrastructure.
 - Astrophysics-relevant commercial hardware, including spacecraft buses that meet the precision needs of astrophysics (e.g., precision pointing) and telescope/telescope assemblies
- Please include demand signals and performance/cost targets that shape your business case.

Topic H: Commercialization Interest in VTA Earth Science and Space Weather Missions

Identify VTA Earth Science mission concepts, demonstrations, and/or Space Weather mission enhancement/augmentations, and/or astrophysics-relevant hardware that could be:

- Competitively provided by commercial services;
- Hosted/integrated on commercial platforms;
- Developed into commercial instrument or data service- product lines;
- Aligned with existing commercial customer segments (public or private).

Indicate desired roles (provider, host, analytics partner), revenue models, and any policy or regulatory enablers needed, especially to facilitate public-private partnerships.

Topic I: Other Relevant Topics

Provide additional insights not captured above. Briefly explain why the topic falls outside established categories.

After the submission of responses, to clarify and better understand the issues raised, NASA may conduct meetings with respondents such as topical workshops. These completely voluntary meetings may provide an opportunity for respondents to provide more detailed feedback to NASA and to clarify any issues as requested by NASA. Any public discussion of the results of this RFI will not disclose the identities of the respondents without their prior approval.

It is not NASA's intent to disclose publicly Respondents' proprietary information obtained in response to this RFI. To the full extent that it is protected pursuant to the Freedom of Information Act and other laws and regulations, information identified by a Respondent as "Proprietary or Confidential" will be kept confidential.

It is emphasized that this RFI is NOT a Request for Proposal, nor is it an Invitation for Bid. This RFI is being used to obtain information for planning purposes only, and the Government does not intend to award a contract at this time. As stipulated in FAR 15.201(e), responses to this notice are not considered offers and cannot be accepted by the Government to form a binding contract. Pursuant to FAR 52.215-3, entitled Request for Information or Solicitation for Planning Purposes, this information is being made available for market research, information, and planning purposes and to allow the broader community the opportunity to verify reasonableness and feasibility of the requirement, as well as promote competition. This RFI is subject to review or cancellation at any time and is not to be construed as a commitment by the Government to enter into a contract. The Government will not pay for the information submitted in response to this request.

Please do not request a copy of the solicitation, as no solicitation exists at this time. If a firm requirement is developed and a solicitation is issued, the solicitation will be made available through NSPIRES (<https://nspires.nasaprs.com/external/>). It is the responsibility of Offerors and interested parties to monitor the internet sites for the release of the solicitation and amendments, if any, and they will be responsible for downloading their own copy of the documents. NASA Clause 1852.215-84, Ombudsman, is applicable. The Center Ombudsman for potential acquisitions can be found at <https://www.hq.nasa.gov/office/procurement/regs/Procurement-Ombuds-Comp-Advocate-Listing.pdf>.

4. Requested Information

NASA also requests and collects a limited set of demographic data from all registered NSPIRES users for the purpose of future analysis of the demographic differences associated with NASA's award and review processes. Demographic items requested include sex, race, ethnicity, and disability status. Submission of these demographic data are voluntary, confidential, and not a

precondition for using NSPIRES or submitting a response to this RFI. If you do not wish to provide demographic responses, choose the buttons labeled “I Prefer Not to Answer”.

Respondents should include:

- Name, organization, and contact information
- Names and affiliations of any co-investigators
- Identification of one primary topic and up to three secondary topics (Section 3)
- Responses addressing the chosen topic(s)

Commercial respondents are encouraged to provide:

- Market segments served and commercial use cases
- Expected commercial demand for VTA relevant- technologies or missions (customers, price points, service levels)
- Integration constraints (interfaces, manufacturability, test requirements), platform compatibility, and schedule/cost drivers
- Any data rights/licensing preferences and IP considerations relevant to collaboration with NASA

Do not submit proprietary, export controlled, or CUI information.

5. Response Instructions

All responses must be received by 11:59 p.m. Eastern time on April 23, 2026.

All responses to this RFI must be submitted in an electronic format via the NASA Solicitation and Proposal Integrated Review and Evaluation System (NSPIRES), located at <http://nspires.nasaprs.com/>.

There is no limit on the number of responses from a particular submitter; that is, an individual, an institution, or organizational units at an institution. Individuals currently unaffiliated with any commercial, research, educational, or other institution may list their institutional affiliation as “Self”. Submissions from outside the United States also are welcome.

For this RFI, a response submission will take the form of a Notice of Intent (NOI) within the NSPIRES online announcement data management system. The RFI response itself will be a PDF-formatted document that is attached (uploaded) to the NSPIRES system. All responses must adhere to the following formatting requirements:

- Length: 5 pages
- Page size: 8.5” x 11.0” paper size
- Paragraph: Single-spaced, single-column text with no more than 5.5 lines per vertical inch of text
- Margins: One-inch margins on all four sides with no content in the margins
- Font: Font size 12, not to exceed 15 characters per horizontal inch, including spaces, sans serif font recommended
- Inserts: Figures, tables, and other inserts are permitted

You must be registered with NSPIRES to submit an RFI response. See registration instructions at <http://nspires.nasaprs.com/> (select “Create an Account” at the top right of the landing page). Neither institution registration nor an institution affiliation is required to respond to this RFI.

When you have an NSPIRES account, go to the NSPIRES page for “Science as a Service”
Number: NNH26ZDA0**L.

1. Click “Create”
2. Log in when requested
3. Enter the Request for Information: Response title NOTE: NSPIRES will show this as “NOI title”
4. Select "do not link at this time" for submitting organization page.
5. Click "Save" on next page.
6. It is not necessary to complete any of the "NOI Details"; all requested information should be included in the attached PDF document. Information which is entered into “NOI Details” but not included in the attached PDF document will not be considered
7. Prepare your RFI response offline and save as a PDF document (NSPIRES instructions on PDF formats are described in the NSPIRES PDF Guidelines available at https://nspires.nasaprs.com/external/externalhelp/public/index.htm#t=NSPIRES_Documents_main.htm). The response document must include the respondent’s Name, institution, and e-mail address so the file is self-contained. File names format should be “PI Last Name - First Name – Number - RFI”. “Number” will be used to distinguish multiple responses from the same PI. The response should not exceed 5 pages in length excluding references.
8. To attach (upload) your PDF document:
 - a.) Click "add" under NOI attachments section;
 - b.) Select "Proposal Document" from the drop-down list;
 - c.) Browse to attach your PDF file;
 - d.) Select "Upload";
 - e.) Click "OK";
 - f.) Your RFI document has been uploaded to NSPIRES.
9. Click Submit NOI button. NOTE: Clicking “Submit NOI” does not complete the submission process.
10. Ignore any warnings about incomplete NOI elements.
11. Click "Submit". This will take you to the NOI submission confirmation page, which provides you with the NOI/RFI number for your records.

Please note: You may delete and replace form fields and upload documents any time before the submission deadline. However, once your RFI is submitted, it cannot be deleted. If you find that you have submitted in final and need to make a correction, then indicate in the title of any full second submission in capital letters: “CORRECTED RESPONSE”, and the first submission will not be reviewed. If you submit more than one application without indication of which is the “Correct” submission, then nothing will be reviewed. Unsubmitted responses will not be considered.

7. Points of Contact

Questions regarding this RFI may be directed to:

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